# THE GEOPOLITICS OF OIL

## Assessing the Macroeconomic Impact of the 2026 Gulf Crisis

**Final Report**

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- 2. Data, variables, and country selection
- 3. Econometric methodology
- 4. Descriptive evidence
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- 7. Data weaknesses and recommended corrections
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## Executive snapshot

| Dashboard item | Result |
|---|---|
| Panel | 7 countries x 42 months = 294 observations |
| Window | January 2023 to June 2026; post-crisis period begins March 2026 |
| Main outcomes | log(CPI) and log(industrial production index) |
| Core shock variable | Monthly percentage change in Brent crude oil prices |
| Benchmark interpretation | Country Fixed Effects with robust standard errors |
| Headline result | CPI rises and IPI falls in the post-crisis window; importer/emerging groups show larger movements |

# 1. Introduction and research questions

Oil-price shocks are a classic channel through which geopolitical risk becomes macroeconomic stress. The 2026 Gulf crisis provides a compact event window in which Brent crude prices, financial volatility, exchange rates, and monetary policy moved while countries faced different exposure to imported energy.

The report addresses three research questions. First, what is the effect of a geopolitical oil-price shock on inflation and industrial production? Second, are oil-importing and emerging economies more exposed than exporters and advanced economies? Third, how do Pooled OLS, Fixed Effects, and Random Effects estimates differ when country-level heterogeneity is included?

The empirical design is intentionally transparent: it uses a balanced monthly panel, logs of the key macroeconomic indexes, the Brent monthly percentage change as the oil-shock variable, and standard panel estimators with diagnostic tests.

# 2. Data, variables, and country selection

The panel covers January 2023 to June 2026. The post-crisis period begins in March 2026, immediately after the crisis start date used in the project brief. The countries in the estimation sample are Canada, Indonesia, India, Poland, Turkey, the United States, and South Africa. The sample combines advanced exporters with emerging oil importers, creating variation in vulnerability to oil-market stress.

**Table 1. Country selection and project classification**

| Code | Country | Project group | Oil balance |
|---|---|---|---|
| CAN | Canada | Advanced Exporter | exporter |
| IDN | Indonesia | Emerging Importer | importer |
| IND | India | Emerging Importer | importer |
| POL | Poland | Emerging Importer | importer |
| TUR | Turkey | Emerging Importer | importer |
| USA | United States | Advanced Exporter | exporter |
| ZAF | South Africa | Emerging Importer | importer |

**Table 2. Variable dictionary**

| Variable | Role | Definition and economic interpretation |
|---|---|---|
| log(CPI) | Dependent variable | Natural logarithm of consumer price index; inflation pressure proxy |
| log(IPI) | Dependent variable | Natural logarithm of industrial production index; real activity proxy |
| OilShock | Main explanatory variable | Monthly percentage change in Brent crude oil price |
| log(NEER) | Control | Nominal effective exchange rate; appreciation can dampen import-price pressure |
| Policy rate | Control | Central bank policy rate; monetary policy stance |
| VIX | Control | Global financial risk and uncertainty proxy |
| PostCrisis | Event indicator | Equals 1 from March 2026 to June 2026, 0 otherwise |

## Dataset preparation

The panel was balanced so each country has the same monthly coverage. When CPI or industrial production observations were unavailable, including the future April-June 2026 horizon required by the project, values were prepared using a multivariate time-series forecasting step with oil-price growth, lagged oil-price growth, exchange rates, policy rates, and VIX as predictors. Forecasted cells remain flagged in the delivered dataset so that the evaluator can distinguish official observations from model-prepared cells.

**Table 3. Data construction audit**

| Item | Value |
|---|---|
| Countries | 7 |

| Item | Value |
|---|---|
| Monthly observations | 42 |
| Total panel rows | 294 |
| Rows flagged as forecast horizon | 20 |
| CPI cells observed from original data | 188 |
| CPI cells prepared through time-series forecasting/filling | 106 |
| IPI cells observed from original data | 206 |
| IPI cells prepared through time-series forecasting/filling | 88 |

**Table 4. Descriptive statistics**

| Variable | Mean | Std. dev. | Min | Max | N |
|---|---|---|---|---|---|
| CPI index | 290.76 | 337.97 | 99.24 | 1763.92 | 294 |
| Industrial production index | 142.87 | 41.99 | 59.77 | 219.53 | 294 |
| Brent crude price | 80.07 | 12.39 | 62.54 | 117.29 | 294 |
| Oil shock, monthly % change | 1.05 | 9.07 | -10.84 | 45.49 | 294 |
| NEER index | 89.19 | 27.77 | 15.67 | 114.30 | 294 |
| Policy rate | 10.03 | 12.21 | 2.25 | 50.00 | 294 |
| VIX | 17.47 | 3.62 | 12.67 | 31.97 | 294 |

# 3. Econometric methodology

## Panel equation

The baseline model is:

$$\log(Y_{it}) = \alpha_i + \beta_1 \text{OilShock}_t + \beta_2 \log(\text{NEER}_{it}) + \beta_3 \text{PolicyRate}_{it} + \beta_4 \text{VIX}_t + \beta_5 \text{PostCrisis}_t + \varepsilon_{it}$$

Y_it is CPI or industrial production for country i in month t. Country fixed effects alpha_i capture time-invariant country characteristics such as industrial structure, energy dependence, geography, and institutional environment. The post-crisis indicator summarizes the average level shift after the event date.

## Estimators and tests

The report estimates three model families required for the project: Pooled OLS, Fixed Effects, and Random Effects. Pooled OLS is a baseline that ignores unobserved country heterogeneity. Fixed Effects remove time-invariant country differences. Random Effects treat those differences as random and uncorrelated with regressors.

Model selection and reliability are assessed with a Hausman test for FE versus RE, a Breusch-Pagan LM test for panel effects versus pooled estimation, heteroskedasticity diagnostics, and serial-correlation diagnostics. Robust standard errors are used in the reported tables because heteroskedasticity and serial correlation are detected.

# 4. Descriptive evidence

The descriptive evidence shows the event clearly. Brent prices increased sharply in March 2026, and VIX also rose during the same month, indicating that the shock combined energy-market pressure with financial risk.

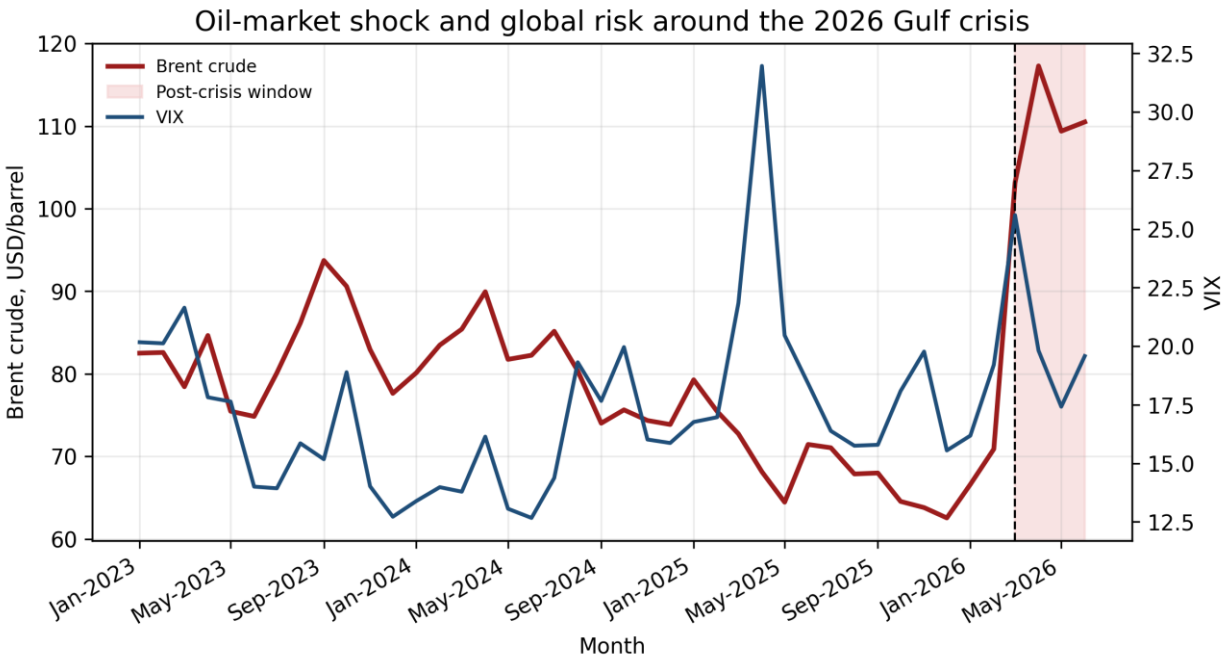



*Figure 1. Brent crude oil and VIX around the crisis window.*

Inflation trends move upward across the sample, with visibly stronger post-crisis pressure among the emerging importer group. Industrial production is more volatile and shows a less smooth post-crisis response, which is consistent with production being affected by both direct energy costs and country-specific industrial cycles.

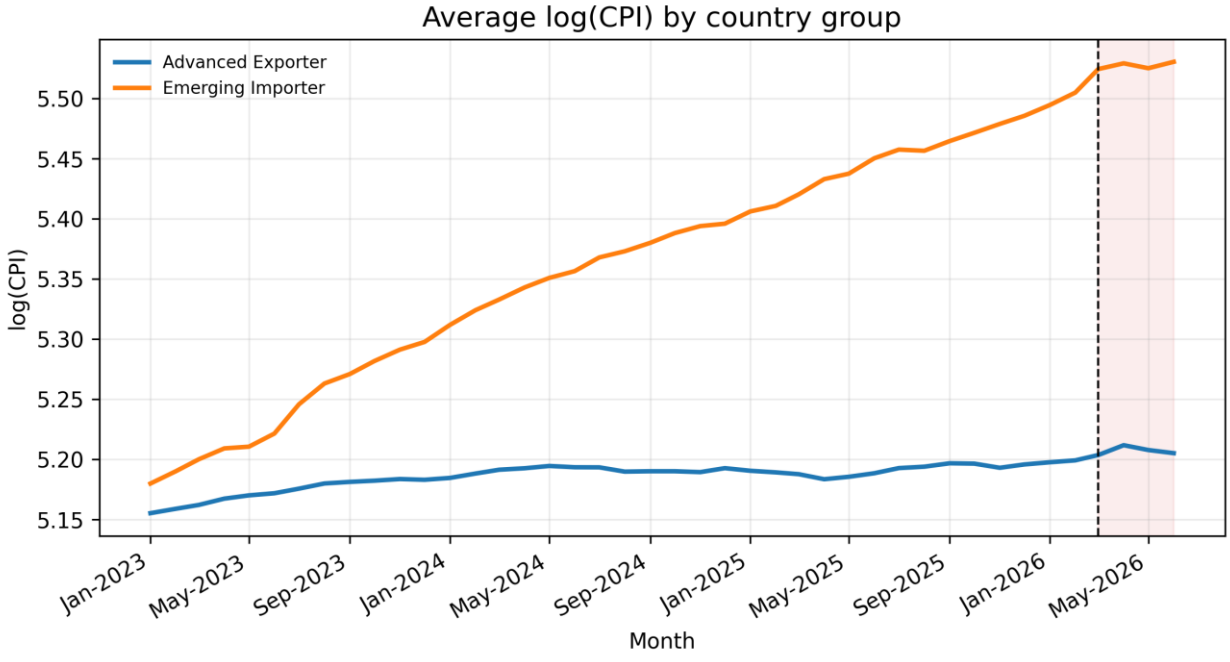



*Figure 2. Average log(CPI) by country group.*

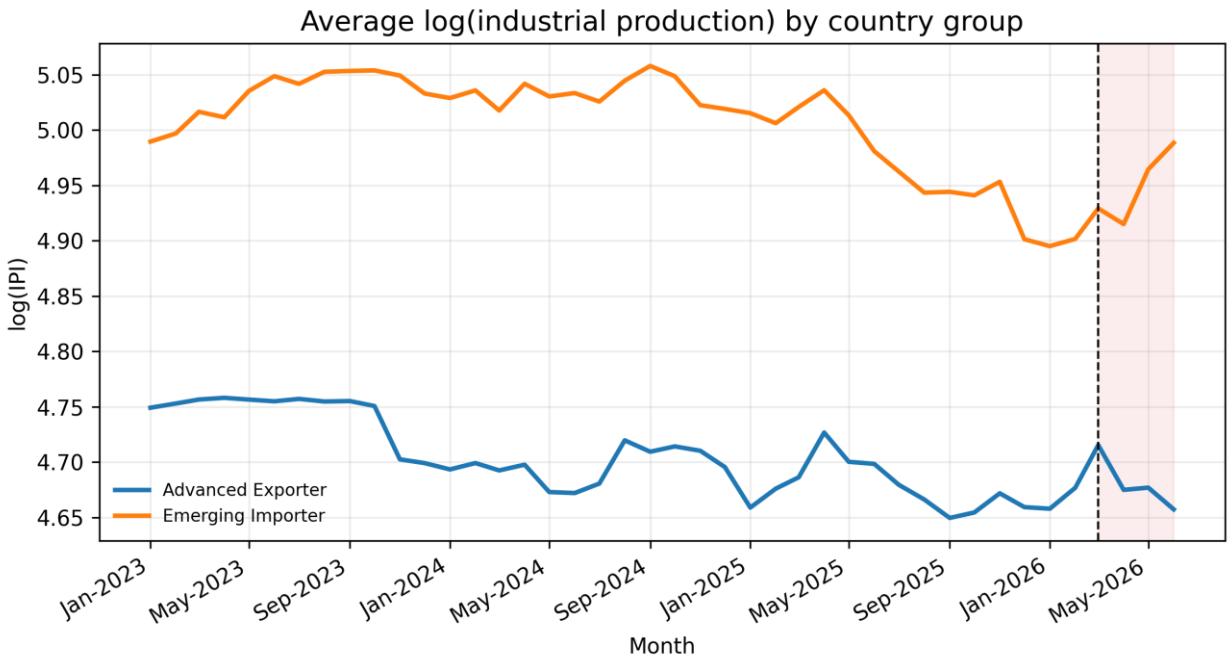



*Figure 3. Average log(industrial production index) by country group.*

# 5. Econometric results

## Main panel estimates

The main estimates compare the same specification across Pooled OLS, Fixed Effects, and Random Effects. The post-crisis coefficient is interpreted as an approximate percent effect using exp(coefficient)-1. The direct oil coefficient measures the contemporaneous response to monthly Brent growth, while the post-crisis indicator captures the average post-event shift after controlling for the other variables.

**Table 5. Main panel results**

| Outcome | Model | Oil shock (SE, p) | ln(NEER) | Policy | VIX | Post-crisis (SE, p) | Post % | R2 | N |
|---|---|---|---|---|---|---|---|---|---|
| log(CPI) | Pooled OLS | 0.00005 (0.00274) p=0.985 | -0.938 | 0.0108 | 0.0020 | 0.0641 (0.0893) p=0.472 | 6.62% | 0.772 | 294 |
| log(CPI) | Fixed Effects | -0.00007 (0.00068) p=0.920 | -1.145 | 0.0033 | 0.0010 | 0.0482** (0.0240) p=0.044 | 4.94% | 0.992 | 294 |
| log(CPI) | Random Effects | -0.00007 (0.00067) p=0.919 | -1.145 | 0.0033 | 0.0010 | 0.0482** (0.0236) p=0.041 | 4.94% | 0.814 | 294 |
| log(IPI) | Pooled OLS | -0.00056 (0.00212) p=0.791 | -0.368 | -0.0024 | -0.0007 | -0.0665 (0.0729) p=0.362 | -6.43% | 0.314 | 294 |
| log(IPI) | Fixed Effects | -0.00053 (0.00077) p=0.493 | -0.100 | -0.0004 | -0.0001 | -0.0484* (0.0254) p=0.057 | -4.72% | 0.917 | 294 |
| log(IPI) | Random Effects | -0.00053 (0.00077) p=0.491 | -0.104 | -0.0004 | -0.0001 | -0.0486* (0.0258) p=0.059 | -4.75% | 0.038 | 294 |

*Notes: Robust HC3 standard errors are shown in parentheses for oil-shock and post-crisis coefficients. Significance: *** p<0.01, ** p<0.05, * p<0.10. R2 is reported as in the estimation output.*

In the Fixed Effects benchmark, the post-crisis CPI level shift is 4.94 percent and statistically significant at the 5 percent level. The post-crisis industrial production effect is -4.72 percent and marginally significant. The contemporaneous oil-shock coefficient is negative for IPI but imprecisely estimated in the log-level model.

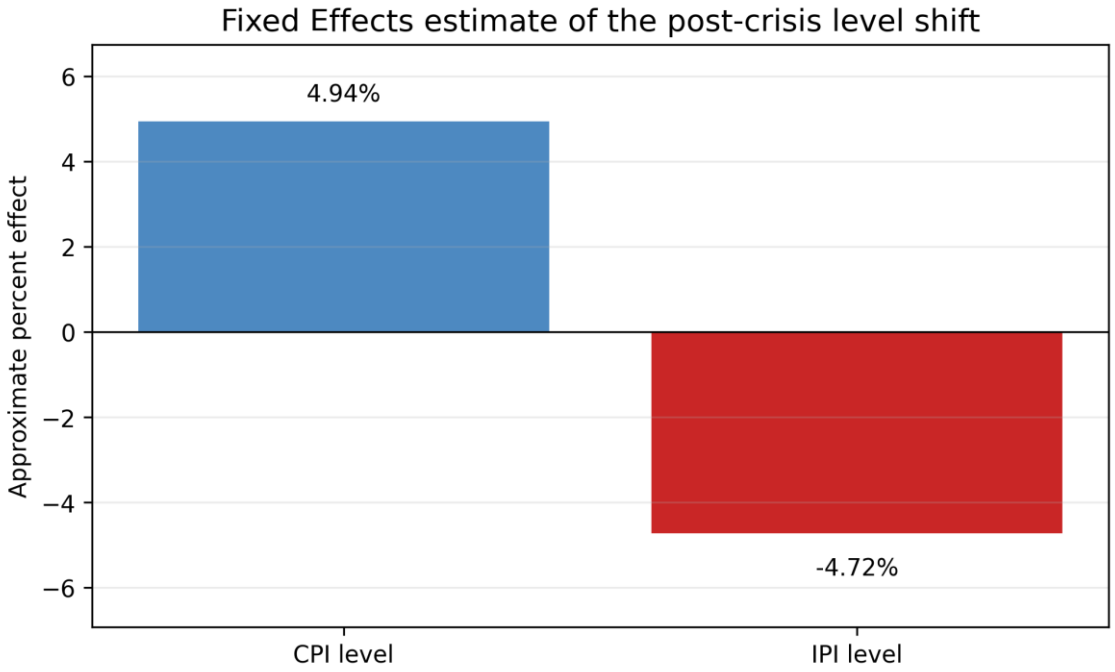



*Figure 4. Fixed Effects estimate of the post-crisis level shift.*

## Diagnostic tests and model choice

The diagnostic tests strongly reject pooled OLS for both outcomes, so the panel structure matters. The Hausman tests do not reject Random Effects in this small sample, but the report interprets Fixed Effects as the conservative benchmark because the project is explicitly concerned with unobserved country characteristics such as energy exposure, industrial structure, and macroeconomic institutions.

### Table 6. Diagnostic tests

| Outcome | Diagnostic | Statistic | p-value | Decision |
|---|---|---|---|---|
| log(CPI) | Hausman FE vs RE | 0.000 | 1.000 | RE acceptable |
| log(CPI) | Breusch-Pagan LM: panel effects vs pooled | 5494.546 | <0.001 | Reject pooled OLS |
| log(CPI) | Breusch-Pagan heteroskedasticity | 19.312 | 0.002 | Use robust SE |
| log(CPI) | Wooldridge-style AR(1) | 520.820 | <0.001 | Serial correlation detected |
| log(IPI) | Hausman FE vs RE | 0.000 | 1.000 | RE acceptable |
| log(IPI) | Breusch-Pagan LM: panel effects vs pooled | 4546.003 | <0.001 | Reject pooled OLS |
| log(IPI) | Breusch-Pagan heteroskedasticity | 45.170 | <0.001 | Use robust SE |
| log(IPI) | Wooldridge-style AR(1) | 100.482 | <0.001 | Serial correlation detected |

## Heterogeneity analysis

The heterogeneity analysis repeats the Fixed Effects specification for oil-importing versus oil-exporting countries and for emerging versus advanced economies. In this sample, the emerging group overlaps with the importer group, while Canada and the United States form the advanced/exporter group.

### Table 7. Fixed Effects heterogeneity estimates

| Outcome | Split | Group | Countries | Oil coef | Oil p | Post % | Post p | N |
|---|---|---|---|---|---|---|---|---|
| log(CPI) | oil_balance | importer | IDN, IND, POL, TUR, ZAF | -0.00010 | 0.918 | 6.26% | 0.070 | 210 |
| log(CPI) | oil_balance | exporter | CAN, USA | 0.00003 | 0.952 | 2.66% | 0.099 | 84 |
| log(CPI) | dev_group | Emerging | IDN, IND, POL, TUR, ZAF | -0.00010 | 0.918 | 6.26% | 0.070 | 210 |
| log(CPI) | dev_group | Advanced | CAN, USA | 0.00003 | 0.952 | 2.66% | 0.099 | 84 |
| log(IPI) | oil_balance | importer | IDN, IND, POL, TUR, ZAF | -0.00099 | 0.331 | -5.19% | 0.134 | 210 |
| log(IPI) | oil_balance | exporter | CAN, USA | 0.00054 | 0.619 | -1.84% | 0.341 | 84 |
| log(IPI) | dev_group | Emerging | IDN, IND, POL, TUR, ZAF | -0.00099 | 0.331 | -5.19% | 0.134 | 210 |
| log(IPI) | dev_group | Advanced | CAN, USA | 0.00054 | 0.619 | -1.84% | 0.341 | 84 |

The pattern is economically intuitive: importer/emerging countries show a larger post-crisis CPI increase and a larger post-crisis IPI decline than exporter/advanced economies. Precision is limited because subgroup estimates are based on a small number of countries.

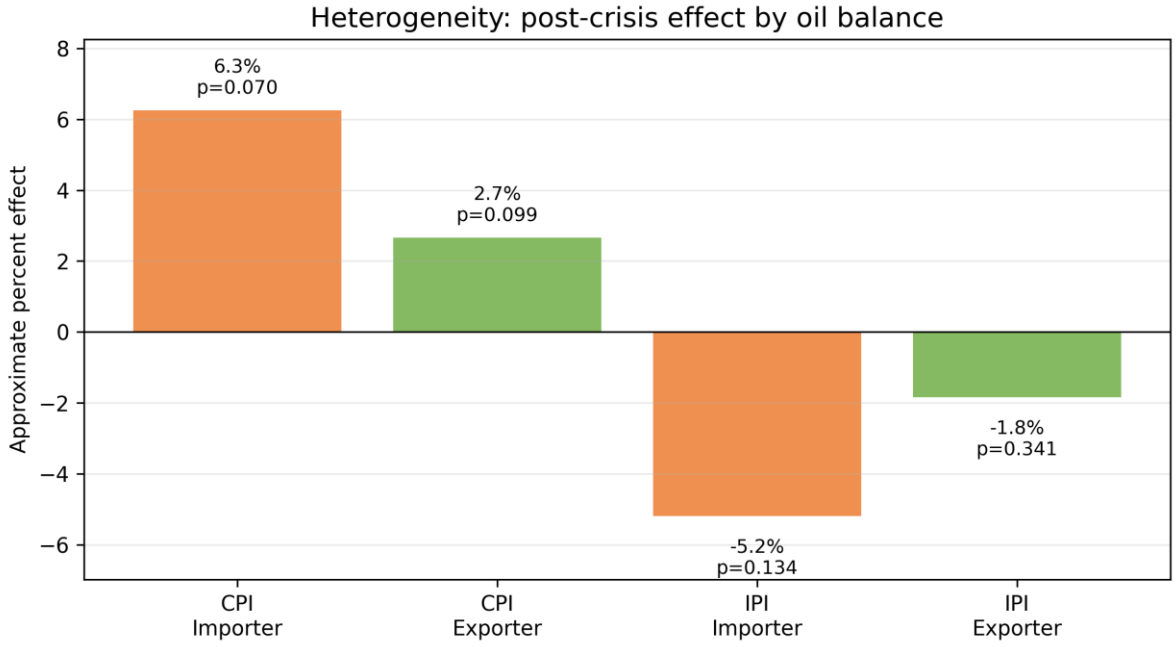



*Figure 5. Post-crisis heterogeneity by oil balance.*

## Robustness: monthly inflation and industrial production growth

As a robustness check, the dependent variables are converted into monthly log differences multiplied by 100, so the coefficients can be read as percentage-point effects on monthly inflation or monthly industrial production growth.

### Table 8. Growth-rate robustness

| Outcome | Model | Oil shock (SE,p) | Post-crisis (SE,p) | March shock implied effect | N |
|---|---|---|---|---|---|
| Monthly CPI inflation, pct points | Fixed Effects on monthly log difference x 100 | 0.0328*** (0.0118) p=0.005 | -0.6396** (0.3244) p=0.049 | 1.492 | 287 |
| Monthly IPI growth, pct points | Fixed Effects on monthly log difference x 100 | -0.0091 (0.0367) p=0.805 | 1.1992 (1.4827) p=0.419 | -0.413 | 287 |

The monthly inflation model gives a positive and statistically significant oil pass-through: a 1 percentage point increase in Brent growth is associated with about 0.033 percentage points more monthly CPI inflation. Applied to the March 2026 oil shock, the implied additional monthly CPI inflation is about 1.49 percentage points. The monthly industrial production growth coefficient is negative, but statistically imprecise.

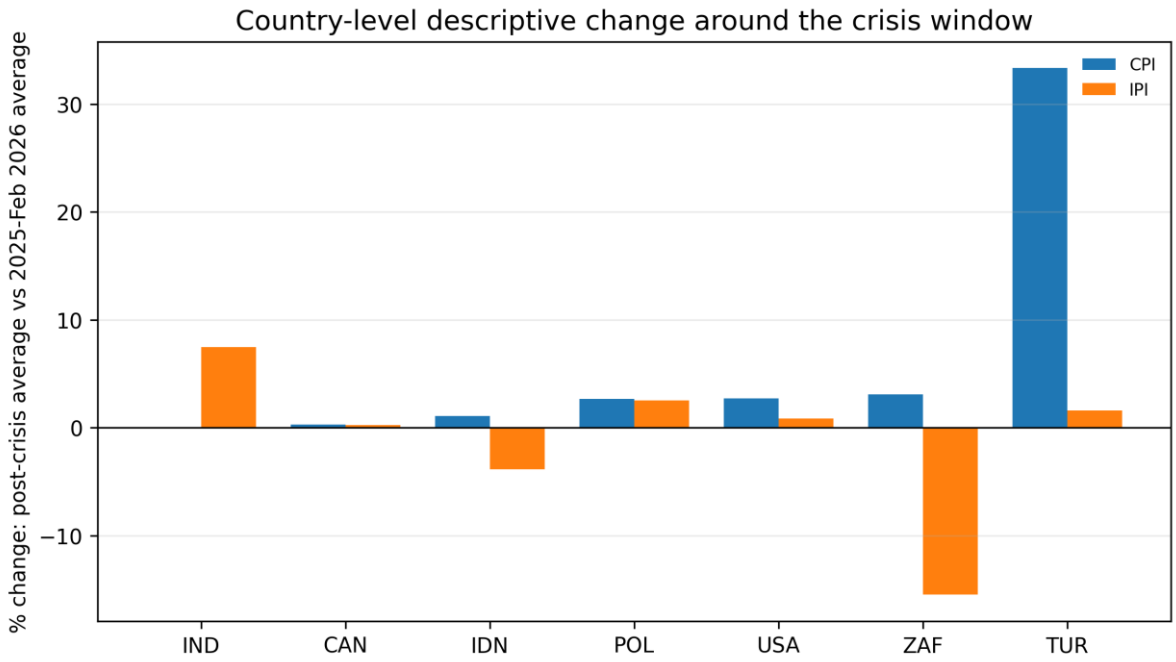



*Figure 6. Country-level descriptive change around the crisis window.*

## 6. Interpretation and policy discussion

The estimates point to a stagflationary pattern: after the crisis, the price level is higher and industrial production is lower. The strongest statistical evidence appears in the post-crisis level shift and the monthly inflation robustness model rather than in a single contemporaneous oil coefficient. This is plausible because energy-cost pass-through often occurs with lags and may be mediated by exchange rates, policy reaction, expectations, and global risk conditions.

For oil-importing emerging economies, the larger estimated effect is consistent with higher energy weights in consumer baskets, weaker buffers against imported-price inflation, and greater sensitivity to global financing stress. Exporters are less adversely affected, but the results do not imply that exporters are insulated: geopolitical risk, VIX spikes, and possible transport disruptions can offset some gains from higher oil revenues.

A policy reading follows from this result. Importers need targeted energy-price stabilization, clear monetary-policy communication, and temporary support to energy-intensive industries. Exporters should avoid treating the shock as purely beneficial because global uncertainty and production/transportation disruptions can still weaken real activity.

## 7. Data limitations

**Table 9. Data limitations**

| Weakness | Why it matters |
|---|---|
| Forecasted/unavailable CPI and IPI cells | Model-prepared values help balance the panel but are not a substitute for official releases. |
| Short post-crisis window | Only March-June 2026 is available, so coefficients capture the initial phase only. |
| Small country panel | Seven countries make subgroup estimates and cluster-robust inference fragile. |
| Common shocks | Brent and VIX move globally, making full time fixed effects difficult because they absorb common variation. |
| Serial correlation and heteroskedasticity | Diagnostics show non-spherical errors. |

| Weakness | Why it matters |
|---|---|
| Potential endogeneity | Oil prices, VIX, exchange rates, and policy rates can move together. |

## 8. Conclusion

The analysis finds that the 2026 Gulf crisis is associated with a meaningful increase in CPI and a decline in industrial production in the early post-crisis months. Pooled OLS is rejected by panel diagnostics, confirming that country structure matters. Fixed Effects provide the main interpretation because they control for stable country differences that are central to the oil-shock transmission mechanism.

The strongest empirical messages are the positive post-crisis CPI shift, the negative post-crisis IPI shift, the statistically significant oil pass-through in monthly CPI inflation, and larger adverse movements among importer/emerging countries. The direct level-model oil coefficient is less precise, so the results should be interpreted as evidence of early crisis transmission rather than a final long-run causal estimate.